

Changes of hepatic myeloid cells in chronic viral hepatitis and after cure and their clinical significance

To the Editor:

Using single-cell RNA sequencing, the article by Cui *et al.* provides an in-depth phenotypic and functional analysis of hepatic myeloid cells from patients with chronic hepatitis C before and after treatment with direct-acting antiviral drugs, which reveals the changes and regulatory mechanisms of hepatic myeloid cells after chronic infection and cure, and provides valuable information for the understanding of hepatic immunity and for the development of new immune therapies.¹ We appreciate the authors' work but have some questions and suggestions.

First, why did the authors choose only patients with hepatitis C for their study and not patients with hepatitis B or other chronic viral hepatitis. Hepatitis B is the most common chronic hepatitis worldwide and one of the major risk factors for liver cancer.² Whether similar heterogeneity and dynamic changes in hepatic myeloid cells exist in patients with hepatitis B, whether they have different phenotypic and functional characteristics from those of patients with hepatitis C, and whether they respond to and are affected by antiviral therapy differently are all questions worth exploring. The authors should compare the characteristics and differences of hepatic myeloid cells in patients with different types of chronic viral hepatitis in future studies to improve the generalizability and clinical significance of the study.

Second, the authors may have overlooked some important cellular markers and functional factors when analyzing subpopulations of hepatic myeloid cells. For example, the authors used only conventional macrophage markers such as CD68 and CD163 when identifying liver macrophages, without taking into account the diversity and plasticity of liver macrophages, as well as the functional differences of different subpopulations.³ Indeed, hepatic macrophages can be divided into two main types: resident Kupffer cells and monocyte-derived macrophages from the peripheral blood. These two types of macrophages are distinctly different in phenotype and function; for example, Kupffer cells mainly express CD68, CD206, and F4/80, whereas monocyte-derived macrophages mainly express CD68, CD163, and CD11b. In addition, liver macrophages can be divided into M1 and M2 types, which are involved in inflammatory responses and tissue repair, respectively. M1-type macrophages predominantly express CD86, CD80, and TNF- α , whereas M2-type macrophages predominantly express CD206, CD163, and IL-10. We suggest that the authors include these cellular markers and functional factors in subsequent analyses to more accurately portray the subpopulation and functional status of liver macrophages, as well as their changes in response to chronic infection and after healing.

Finally, hepatic myeloid cells may play an important role in the development and regression of chronic viral hepatitis, and thus may serve as potential therapeutic targets or prognostic indicators. The authors' study showed that hepatic myeloid cells showed significant phenotypic and functional changes after chronic infection and cure, including downregulation of interferon-stimulated genes, upregulation of MHC-II, expression of PD-L1/L2, and expression of IDO1, which may reflect the role of hepatic myeloid cells in antiviral, immunomodulatory, and tissue repair.¹ Indeed, several clinical studies have found that the number and activity of hepatic myeloid cells are associated with the severity and prognosis of chronic viral hepatitis. For example, neutrophils and macrophages in the liver can release pro-inflammatory factors and reactive oxygen species, leading to liver inflammation and fibrosis. Myelosuppressive cells in the liver, including neutrophils, monocytes, and macrophages, can inhibit T-cell activation and proliferation through the expression of molecules such as PD-L1/L2, IDO1, and other molecules, thereby affecting viral clearance and autoimmunity. Dendritic cells in the liver can enhance antiviral immunity by expressing molecules such as MHC-II, which promotes T-cell differentiation and memory. Thus, hepatic myeloid cells are important players and regulators of chronic viral hepatitis, and can be used as biomarkers for assessing the disease and prognosis, as well as potential targets for improving therapeutic outcomes and preventing complications.

In summary, we commend Cui *et al.* for their study and hope that they will further refine it.

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Authors' contributions

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Supplementary data

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